

Derivation and validation of the AJCC9V pathological stage classification for HPV-positive oropharyngeal carcinoma: a multicentre registry analysis



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Summary

Background Modernisation of the American Joint Committee on Cancer and Union for International Cancer Control (AJCC/UICC) staging for HPV-positive oropharyngeal carcinoma has strengthened its fundamental role to convey prognosis and guide treatment decisions. Implementation has nonetheless revealed imbalances. An AJCC committee reappraised HPV-positive oropharyngeal carcinoma pathological staging to advance prognostic accuracy and improve clinical applicability.

Methods For this analysis, US registry data from the National Cancer Database were divided into derivation and validation cohorts. Eligible cases encompassed surgically treated adult patients aged 18 years or older with HPV-positive oropharyngeal carcinoma. The primary objective was to derive and validate an optimised HPV-positive oropharyngeal carcinoma pathological staging classification based on determinants of overall survival. Multivariable Cox regression models were used to assess lymph node characteristics for overall survival. Non-linear associations between metastatic lymph node number and survival were modelled with restricted cubic splines. Adjusted hazard ratios (AHRs) and recursive partitioning analysis methods were applied to generate optimal classification groups. Performance was evaluated with Groome's criteria.

Findings Overall, 14 447 patients across 984 facilities in the USA met the criteria for inclusion, and were divided into a derivation cohort (n=7768) and validation cohort (n=6679). Patients were treated between 2010 and 2019; 12 276 (85.0%) were male, 2171 (15.0%) were female, 13 594 (94.1%) were White, and 4552 patients (31.5%) had pathological extranodal extension (pENE). Median follow-up was 52.4 months (95% CI 51.5–53.3). Mortality risk increased with each additional metastatic lymph node (HR 1.20 [95% CI 1.11–1.29], $p<0.0001$), up to an optimal cutoff at 4.3 lymph nodes. Multivariable analysis confirmed the association of pENE with increased mortality risk (HR 1.47 [95% CI 1.30–1.65], $p<0.0001$), but no significant prognostic changes were shown by the extent of pENE (minor vs major). AHR approaches derived and validated optimised pN and pTNM stage categories (N1a: 1 positive lymph node and pENE-negative; N1b: 2–4 positive lymph nodes and pENE-negative; N2: >4 positive lymph nodes and pENE-negative or 1–4 positive lymph nodes and ENE-positive; N3: >4 positive lymph nodes and ENE-positive; Stage I: T0-2N0-1M0; Stage II: T0-2N2-3M0 or T3N0-2M0; Stage III: T3N3M0 or T4N0-3M0; Stage IV: M1. AJCC9V showed superior hazard consistency, outcome prediction, and balance, but not hazard discrimination, compared to AJCC8E. The AJCC Expert Panel on HPV-positive oropharyngeal carcinoma endorsed the proposal by Delphi consensus.

Interpretation The AJCC9V HPV-positive oropharyngeal carcinoma staging classification confers an improved schema for guiding prognostication and management compared to AJCC8E. Incorporating ENE and correcting imbalances will enhance clinical relevance and align pathological staging in a condition whose management continues to evolve.

Funding None.

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Introduction

The emergence of human papillomavirus (HPV)-positive oropharyngeal carcinoma has reshaped head and neck cancer management. Its growing prevalence has markedly outpaced that of other head and neck cancers, accounting for 45% of all head and neck malignancies and exceeding 80% of all oropharyngeal carcinoma cases in the USA.^{1,2} Among HPV-related malignancies, the

incidence of HPV-positive oropharyngeal carcinoma has now overtaken that of cervical cancer in several countries.³ In turn, the viral-driven biology of HPV-positive oropharyngeal carcinoma imparts a comparative mortality risk two to three times lower than that of smoking-driven HPV-negative oropharyngeal carcinoma.⁴ Arguably, some of the most innovative trials in head and neck cancer currently involve de-intensifying

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Research in context

Evidence before this study

We searched PubMed and Medline between Jan 1, 2010, and Jan 1, 2025, with the keywords “HPV”, “oropharyngeal cancer”, and “staging.” HPV-positive oropharyngeal carcinoma is established as a major, if not unique, disease within the spectrum of head and neck malignancies. It now represents nearly half of all head and neck cancers in the USA and confers a conspicuously superior prognosis compared to HPV-negative oropharyngeal carcinoma. Active investigation is underway to ascertain optimal de-intensified management, and in 2018 a separate American Joint Committee on Cancer and Union for International Cancer Control (AJCC/UICC) staging system (AJCC8E) was created to reflect the distinct, virus-driven biology of this disease. However, implementation of AJCC8E has uncovered imbalances that warrant refinement. This includes disproportionate representation of early-stage disease, a disconnect between staging and management, and the absence of controversial prognostic factors such as pathological extranodal extension that have become widely adopted in ongoing trials and clinical decision making.

Added value of this study

An international AJCC expert panel reappraised HPV-positive oropharyngeal carcinoma pathological staging. Leveraging a large-scale analysis of 14 447 surgical cases divided into training (west, midwest) and validation (east, south) cohorts, multivariable analysis identified metastatic lymph node number and pathological extranodal extension as the strongest nodal covariates that correlate with survival. An adjusted hazard ratio approach generated proposed N classification and staging schema integrating these covariates, which were then validated. Version 9 of the AJCC staging system (AJCC9V) was shown by Groome’s criteria to show improved comparative performance.

Implications of all the available evidence

Through refinement of nodal classification and integration of pathological extranodal extension, AJCC9V will improve hazard consistency within each stage, achieve greater balance between groups, and better align staging with treatment patterns.

HPV-positive oropharyngeal carcinoma treatment to reduce toxicities while preserving outcomes.^{5,6}

Oropharyngeal carcinoma staging underwent major structural changes in the 8th edition of the American Joint Committee on Cancer stage classification (AJCC8E). Such shifts included recognising HPV-positive oropharyngeal carcinoma as a distinct entity worthy of its own staging schema; streamlining nodal determinants of prognosis; and reserving Stage IV exclusively for M1 disease. With the growing use of transoral robotic surgery, AJCC8E recognised the unique characteristics of patients with surgically managed HPV-positive oropharyngeal carcinoma and implemented separate cTNM and pTNM systems. This break from the HPV-negative staging convention restored hazard discrimination between stages. It also struck a practical balance between complexity, relevance, and usability—all influential elements that govern staging principles.

While AJCC8E advanced HPV-positive oropharyngeal carcinoma staging, it nonetheless harbours imbalances.⁷ Distribution is one-sided, both for pN classification (>85% of cases are N1) and stage groupings (>80% of cases are Stage I). As such, within these dominant groupings wide variance in hazard exists: too many Stage I cases confer a disconnect between staging and risk stratification. By extension, staging and treatment have become less coordinated: some T1N1 patients undergo observation after surgery, others warrant radiotherapy, while others receive chemoradiotherapy. Finally, prognostic features that have gained prominence in trial design and clinical practice—such as extranodal extension (ENE)—are absent and justify re-evaluation.

Updated data in the form of prospective studies and maturing nationwide registries support refinement for HPV-positive oropharyngeal carcinoma staging. Here, an AJCC working group reappraised the current system via large-scale analysis, dividing cases into two cohorts to derive new schema and confer validation.

Methods

Study design

An AJCC Expert Panel on HPV-positive oropharyngeal carcinoma was convened, and a literature review was conducted to identify areas of improvement. The panel met over 4 months with three rounds of data iteration to achieve consensus. Feedback was provided by the AJCC Executive Committee, Union for International Cancer Control (UICC), and American Society of Clinical Oncology (ASCO) Evidence Based Medicine Committee (EBMC), with key clinical considerations incorporated. All committees approved the final schema.

Data were abstracted from the National Cancer Database (NCDB), a tumour registry maintained by the American Cancer Society and the American College of Surgeons. No patient representatives were involved in the study. The study was deemed exempt from review by the Cedars-Sinai institutional review board. All patients aged older than 18 years who underwent upfront surgical resection with neck dissection (minimum ten lymph nodes examined) for HPV-positive oropharyngeal carcinoma for curative intent were eligible for inclusion (appendix p 2). Data were dichotomised into derivation (west, midwest) and validation (east, south) cohorts.

ENE definitions have undergone changes over time within NCDB. From 2010 to 2017, NCDB encoded pENE

See Online for appendix

as pENE-negative, pENE-positive microscopic, pENE-positive macroscopic, or pENE-positive NOS (not otherwise specified). Microscopic was defined as any pENE mentioned in the microscopic section of the pathology report, while macroscopic was defined as any pENE mentioned in the gross section. From 2018 to 2019, the NCDB defined microscopic as 2 mm or smaller and macroscopic as greater than 2 mm, but also allowed coders to input exact pENE measurements (in mm). We recognise that investigators might use other terms (major or minor) or define microscopic differently (eg, <1 mm). For the main analysis, we described pENE-positive as any pENE positivity (encompassing 2010–19) to be most inclusive. For subset analyses on pENE degree, we specified pENE-positive minor as up to 2 mm and pENE-positive major as greater than 2 mm (encompassing 2018–19).

Procedures

We addressed potential sources of bias using strict inclusion criteria and performing multivariable analyses adjusting for potential confounders. Sex and race were defined via extraction of electronic medical records.

Outcomes

The primary objective was to derive and validate an HPV-positive oropharyngeal carcinoma pathological staging classification based on determinants of overall survival. Adjusted hazard ratio (AHR) data in conjunction with clinical considerations were used to formulate an optimal staging schema.

Statistical analysis

We compared covariates using the Wilcoxon rank sum test for continuous variables and Pearson's χ^2 test for categorical variables (appendix pp 9–10). Median follow-up time was estimated with the reverse Kaplan–Meier method. We used a multivariable Cox proportional hazards regression model to assess the effect of binary pENE involvement (pENE-positive vs pENE-negative) and the degree of pENE involvement (pENE-positive minor vs pENE-positive major vs pENE-negative) on overall survival. To address potential bias due to missing data, we used the multivariate imputation by chained equations (MICE) algorithm.⁸ The imputation process was conducted with 20 iterations to ensure stability of estimates. The proportional hazards assumption was verified with Schoenfeld residuals to ensure model validity.⁹

To assess the effect of increasing metastatic lymph node number on overall survival, we used a restricted cubic spline function to model non-linear relationships, with knot placement at one, two, and eight lymph nodes.¹⁰ The change point in the log relative hazard was assessed with piecewise linear regression, enabling the identification of significant shifts in hazard with increasing metastatic lymph node number.

AHRs and recursive partitioning analysis (RPA) approaches based on a multivariable Cox model were used to inform nodal classification.¹¹ We estimated AHRs for mortality risk for ENE (ENE-positive, ENE-negative) and metastatic lymph node combinations (1 positive lymph node, 2–4 positive lymph nodes, >4 positive lymph nodes) in the derivation cohort, adjusting for NCDB covariates to derive N1–N3 schema. This was then tested for fidelity on the validation cohort. We separately performed an RPA by building a conditional inference tree and identifying binary partitions of prognostic factors (ENE and metastatic lymph node number) that stratified patients into subgroups based on survival. The conditional inference tree was based on a permutation test of a log-rank statistic with Bonferroni-adjusted p values for multiple comparisons.

For staging classification, we similarly computed AHRs for mortality risk for all T and N combinations within the derivation cohort, adjusting for NCDB covariates to derive Stage I–III schema. This was then tested for fidelity on a validation cohort. Cross-tabulation was performed by generating frequencies of each AJCC8E stage and assessing how these were stratified by AJCC9V stage.

AJCC9V was compared to AJCC8E by use of Groome's criteria¹² with the remss package.¹³ This framework provides an objective evaluation of cancer stage classifications by assessing consistency of outcomes for patients within a given stage (hazard consistency), separation of survival outcomes between stages (hazard discrimination), model fit improvement, variance explanation, and patient distribution (balance between stages; appendix p 2).

All statistical analyses were conducted with R software (version 4.4.2). Statistical significance was determined by two-sided p values less than 0.05.

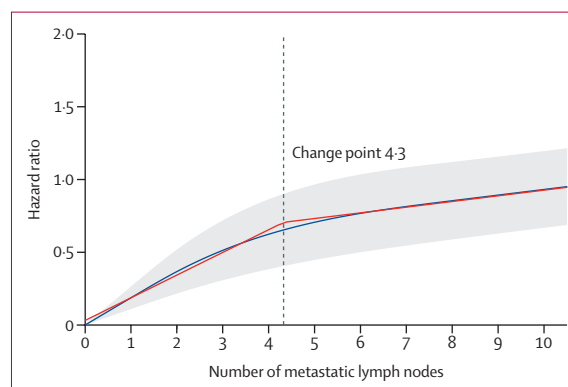


Figure 1: Adjusted mortality risk with increasing number of metastatic lymph nodes in HPV-positive oropharyngeal carcinoma

Using derivation data, regional metastatic lymph node was fit with a multivariable restricted cubic spline plot, illustrating adjusted hazard ratios with increasing number of metastatic lymph nodes for human papillomavirus (HPV)-positive oropharyngeal carcinoma. Three knots were placed at 1, 2, and 8 lymph nodes. The estimated change point was 4.3 metastatic lymph nodes.

Role of the funding source

There was no funding source for this study.

Results

Overall, 14447 patients with surgically treated HPV-positive oropharyngeal carcinoma across 984 facilities between 2010 and 2019 met the inclusion criteria, divided into a derivation cohort (7768 patients across 527 facilities) and validation cohort (6679 patients across 457 facilities; appendix pp 3, 7–8). 12276 (85.0%) patients were male and 2171 (15.0%) were female. 13594 (94.1%) were White, 495 (3.4%) were Black, and 358 (2.5%) were of another race or ethnicity. Median follow-up was 52.4 months (95% CI 51.5–53.3). The median number of lymph nodes examined per case was 31 (IQR 22–42). The mean number of positive metastatic nodes was 2.5 (SD 3.6), and 31.5% (4552 of 14447) showed pENE. On multivariable analysis among non-nodal covariates, advanced T classification such as

T4 (HR 2.86 [95% CI 2.32–3.52], $p < 0.0001$), positive margins (1.29 [1.14–1.45], $p < 0.0001$), and lymphovascular invasion (LVI; 1.35 [1.20–1.50], $p < 0.0001$) were independently associated with increased mortality risk (appendix p 9).

An increasing number of metastatic lymph nodes was significantly associated with worse overall survival on multivariable analysis (appendix p 4). Metastatic lymph node size was not associated with mortality (appendix p 9). On restricted cubic spline analysis, mortality risk escalated continuously with increasing number of metastatic nodes (figure 1). Mortality risk increased (HR 1.20 [95% CI 1.11–1.29]; $p < 0.0001$) with each additional metastatic lymph node up to an identified changepoint of 4.3 lymph nodes, and increased further (HR 1.04 [95% CI 1.02–1.06], $p < 0.0001$) with each additional metastatic lymph node beyond this point.

Overall, pENE-negative patients had improved outcomes compared with pENE-positive patients (3-year overall survival 94.8% [95% CI 94.3–95.3] vs 89.7% [88.8–90.7], $p < 0.0001$). On multivariable analysis, the presence of pENE was found to be independently associated with increased mortality risk (HR 1.47 [95% CI 1.30–1.65], $p < 0.0001$; table 1, appendix pp 9–10).

On evaluating pENE degree or extent, only 2 years of data (2018–19) were available ($n=4767$). There were mixed 3-year overall survival outcomes for pENE-negative (93.6% [95% CI 92.5–94.8]), pENE-positive minor (95.6% [93.7–97.7]), and pENE-positive major (89.7% [85.9–93.7]) cases ($p=0.042$; appendix p 4). On multivariable analysis, pENE-positive minor (HR 0.99 [95% CI 0.63–1.55], $p=0.96$) and pENE-positive major (1.38 [0.85–2.26], $p=0.20$) cases did not show significant prognostic changes (table 1, appendix pp 11–12).

Based on the cubic spline results and AJCC8E precedent, a cutoff of four positive lymph nodes served as the foundation for N categories (1–4 positive lymph nodes, >4 positive lymph nodes). Cases with one positive lymph node and ENE-negative status were then further considered. As they represent a large proportion of cases and because many patients undergo single-modality treatment, these cases were defined as their own category, leading to three groups (1 positive lymph node, 2–4 positive lymph nodes, and >4 positive lymph nodes).

An AHR analysis of the derivation cohort was then performed to integrate ENE status, with stepwise hazard ranges established (figure 2). This led to the creation of N1a (1 positive lymph node and ENE-negative; AHR 1.00–1.35), N1b (2–4 positive lymph nodes and ENE-negative; AHR 1.36–1.45), N2 (>4 positive lymph nodes and ENE-negative or 1–4 positive lymph nodes and ENE-positive; AHR 1.46–2.10), and N3 (>4 positive lymph nodes and ENE-positive; AHR >2.10) classifications (table 2). Analysis of the validation cohort led to similar findings: N1a (AHR 1.00–1.30), N1b

	pENE status (2010–19)		pENE extent (2018–19)	
	HR (95% CI)	p value	HR (95% CI)	p value
Age	1.03 (1.02–1.04)	<0.0001	1.04 (1.02–1.05)	<0.0001
Sex				
Male	..	..	..	..
Female	0.89 (0.77–1.03)	0.11	0.85 (0.60–1.21)	0.37
T stage				
T1	..	..	..	..
T2	1.29 (1.16–1.44)	<0.0001	1.00 (0.75–1.33)	0.98
T3	2.01 (1.69–2.39)	<0.0001	2.14 (1.40–3.29)	0.0005
T4	2.86 (2.32–3.52)	<0.0001	3.66 (2.13–6.29)	<0.0001
Number of metastatic lymph nodes	1.05 (1.04–1.05)	<0.0001	1.06 (1.04–1.08)	<0.0001
Margins				
Negative	..	..	..	..
Positive	1.29 (1.14–1.45)	<0.0001	1.57 (1.15–2.15)	0.0040
Postoperative radiotherapy				
No	..	..	..	..
Yes	0.66 (0.58–0.76)	<0.0001	0.55 (0.41–0.74)	<0.0001
Postoperative chemotherapy				
No	..	..	..	..
Yes	0.99 (0.87–1.13)	0.89	0.74 (0.51–1.08)	0.12
pENE status				
Negative	..	..	..	..
Positive	1.47 (1.30–1.65)	<0.0001	..	..
pENE extent				
Negative	..	..	..	..
Positive minor	..	..	0.99 (0.63–1.55)	0.96
Positive major	..	..	1.38 (0.85–2.26)	0.20

Multivariable Cox regression for overall survival stratified by pENE status (positive vs negative; 2010–19) and pENE extent (negative vs positive minor vs positive major; 2018–19). Covariates in the full models (appendix pp 9–12) included age, sex, race, tumour site, facility type, region, insurance status, median zip code income, proportion of adults in zip code with high school diploma, distance traveled to facility, Charlson-Deyo comorbidity score, T classification, number of lymph nodes examined, number of metastatic lymph nodes, metastatic lymph node size, lymphovascular invasion, pENE status, margins, postoperative radiotherapy, and postoperative chemotherapy. HPV=human papillomavirus. HR=hazard ratio. pENE=pathological extranodal extension.

Table 1: Abridged multivariable analysis of overall survival for HPV-positive oropharyngeal carcinoma

(AHR 1.31–1.50), N2 (AHR 1.51–2.50) and N3 (AHR >2.50; figure 2). One discrepancy entailed the 1 positive lymph node and ENE-positive group (AHR 1.46), which fell marginally outside its expected N2 range (AHR 1.50–2.50) and performed better than the 2–4 positive lymph nodes and ENE-negative group (AHR 1.54; N1b) in the validation cohort. Given that its AHR value approximated the expected range and that patients who have one positive lymph node and are ENE-positive are generally treated with postoperative chemoradiotherapy, whereas patients in the N1b group (2–4 positive lymph nodes and ENE-negative) generally are not, it was decided to maintain the 1 positive lymph node and ENE-positive classification as N2 for clinical practicality. The merged hazard was furthermore congruent with stepwise escalation (appendix p 13). Separately, an independent RPA identified multiple distinct clusters based on ENE and metastatic lymph node number. Grouping of the clusters led to a nodal schema that mirrored and supported the AHR findings (appendix p 5).

Concern was raised by some panel members that the new N classification schema might introduce complexity without meaningfully improving prognostication. It was also suggested, based on data from the ECOG-ACRIN Cancer Research Group Trial (E3311),⁵ that grouping ENE-positive cases less than 1 mm with all other ENE-positive cases (pN2) might imply overtreatment in this subgroup with chemoradiotherapy, when data remain conflicting. Ultimately, the AJCC Expert Panel achieved Delphi consensus (12 of 13 panel members) with this classification.

For patient distribution, N categories in AJCC8E (5398 [86.0%] of 6274 in N1 vs 876 [14.0%] of 6274 in N2) became more broadly balanced in AJCC9V (1645 [29.7%] of 5530 in N1a vs 1313 [23.7%] of 5530 in N1b vs 1945 [35.2%] of 5530 in N2 vs 627 [11.3%] of 5530 in N3; figure 2). This was achieved while preserving separation of curves (5-year overall survival: 93.0% [95% CI 91.5–94.6] in N1a vs 89.3% [87.3–91.3] in N1b vs 86.7% [85.0–88.4] in N2 vs 75.2% [71.5–79.2] in N3; $p<0.0001$).

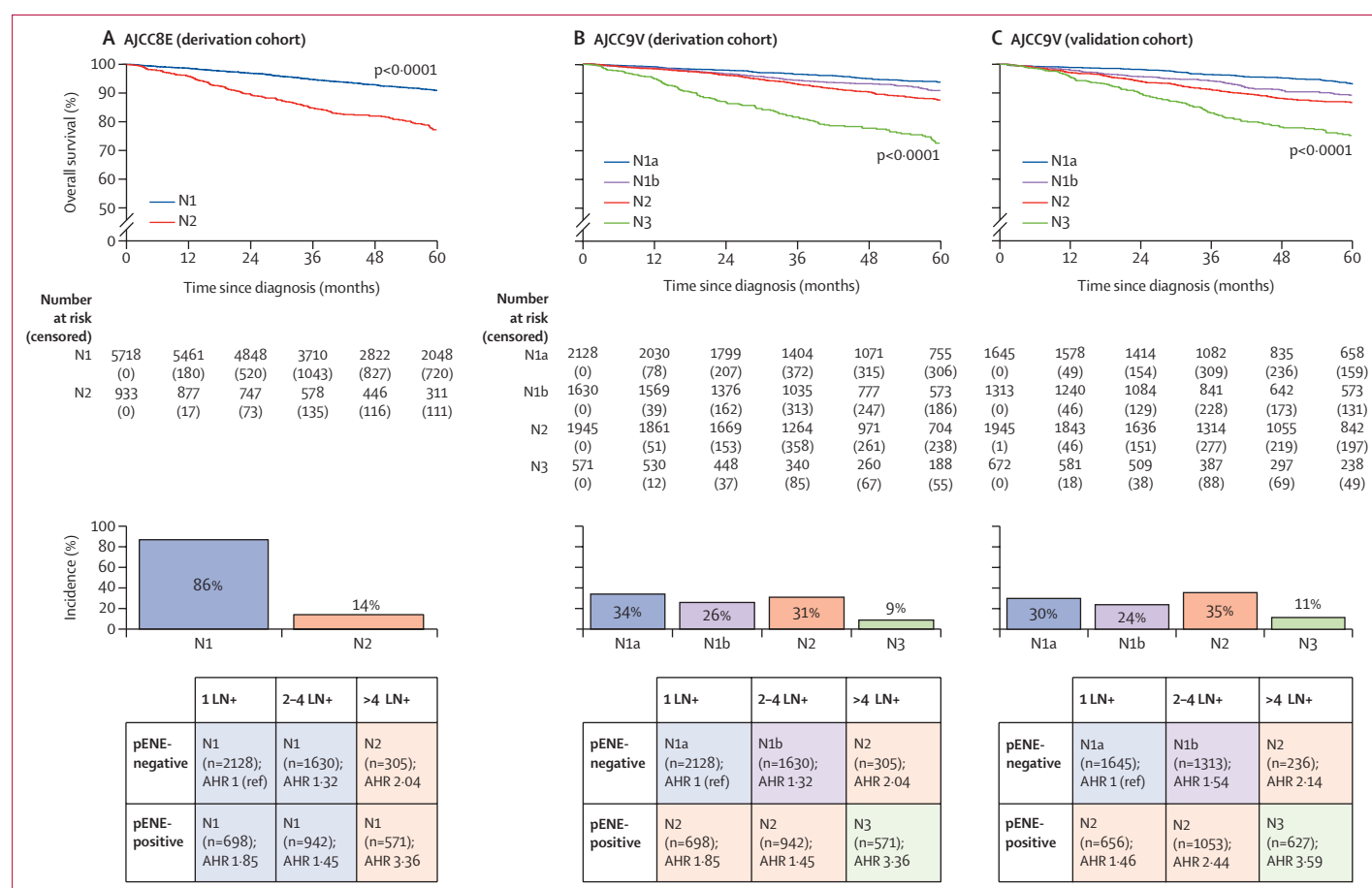


Figure 2: Generation of proposed AJCC9V pathological N classification for HPV-positive oropharyngeal carcinoma with Kaplan-Meier overall survival curves

(A) AJCC8E pN classification (derivation cohort). (B) Proposed AJCC9V pN classification (derivation cohort). (C) Proposed AJCC9V pN classification (validation cohort). AHR=adjusted hazard ratio. AJCC7E=American Joint Committee on Cancer staging system, 7th edition. AJCC8E=American Joint Committee on Cancer staging system, 8th edition. AJCC9V=American Joint Committee on Cancer staging system, version 9. ENE=extranodal extension. HPV=human papillomavirus. LN+=positive lymph node.

	AJCC7E	AJCC8E	AJCC9V
pN classification			
N0	0 positive lymph nodes	0 positive lymph nodes	0 positive lymph nodes
N1	1 positive lymph node (≤ 3 cm)	1–4 positive lymph nodes	..
N1a	..	..	1 positive lymph node and ENE-negative
N1b	..	..	2–4 positive lymph nodes and ENE-negative
N2	..	>4 positive lymph nodes	1–4 positive lymph nodes and ENE-positive >4 positive lymph nodes and ENE-negative
N2a	1 positive lymph node (3–6 cm)	..	..
N2b	>1 positive lymph nodes (≤ 6 cm)	..	..
N2c	>1 positive lymph nodes (bilateral or contralateral; ≤ 6 cm)	..	..
N3	≥ 1 positive lymph nodes (>6 cm)	..	>4 positive lymph nodes and ENE-positive
pTNM stage			
Stage I	T1N0M0	T0-2N0-1M0	T0-2N0-1M0
Stage II	T2N0M0	T0-2N2M0 T3-4N0-1M0	T0-2N2-3M0 T3N0-2M0
Stage III	T3N0M0 T1-3N1M0	T3-4N2M0	T3N3M0 T4N0-3M0
Stage IV	..	M1	M1
Stage IVA	T4aN0-2M0 T1-3N2M0	..	..
Stage IVB	T4bM0 T1-T4N3M0	..	..
Stage IVC	M1	..	..

AJCC7E=American Joint Committee on Cancer staging system, 7th edition. AJCC8E=American Joint Committee on Cancer staging system, 8th edition. AJCC9V=American Joint Committee on Cancer staging system, version 9. ENE=extranodal extension. HPV=human papillomavirus.

Table 2: Comparison of pathological N classification and TNM stage for HPV-positive oropharyngeal carcinoma

Based on the new N classification, we evaluated prognosis according to respective TNM stage groups. Stage IV was reserved for M1 disease as established in AJCC8E. Using the same AHR methodology adjusting for covariates, thresholds were established to categorise three prognostic groups in the derivation cohort: Stage I (T0-2N1; AHR 0.70–1.30), Stage II (T0-2N2-3, T3N0-2; AHR 1.31–2.90), and Stage III (T3N3, T4N0-3; AHR >2.90). This grouping largely concurred on validation, with relative AHR groupings upheld: Stage I (AHR 1.00–1.60), Stage II (AHR 1.61–3.60), and Stage III (AHR >3.60; figure 3).

Small discrepancies were noted. In the derivation cohort, T0-1N2 cases showed an AHR of 1.1 (outside Stage II parameters of AHR 1.3–2.9), but fit appropriately into Stage II in the validation cohort (AHR 1.7). In the validation cohort, T2N1b cases showed an AHR of 2.2 (outside Stage I parameters of AHR 1.0–1.6), but fit appropriately into Stage I in the derivation cohort (AHR 1.3). Finally, T4N1 cases (AHR 1.9–2.3) in the derivation cohort fit outside their assigned Stage III grouping (AHR >2.9), as did T4N1a cases (AHR 3.0) in the validation cohort (AHR >3.6). These findings are likely to be due to small numbers, as few T4 cases undergo surgery in practice. As such, these cohorts were maintained in their respective stage groupings for clinical consistency (table 2). The merged hazard was

found to show appropriate stepwise escalation by stage (appendix p 13).

For distribution of patients, stage groupings in AJCC8E (6327 [81.4%] of 7768 in Stage I vs 1162 [15.0%] of 7768 in Stage II vs 279 [3.6%] of 7768 in Stage III) became more broadly balanced in AJCC9V (3670 [54.9%] of 6679 in Stage I vs 2758 [41.3%] of 6679 in Stage II vs 251 [3.8%] of 6679 in Stage III; figure 3). Cross-tabulation showed reasonable redistribution (appendix p 14). This was achieved while preserving separation of curves (5-year overall survival: 92.3% [95% CI 91.3–93.2] in Stage I vs 85.2% [83.6–86.9] in Stage II vs 71.3% [64.9–78.4] in Stage III; $p < 0.001$).

AJCC9V Stage I (T0-T2N0-N1) was further investigated, given the priority to better align treatment with staging. N1a cases in this cohort (1 positive lymph node and ENE-negative) are often treated with surgery alone, while N1b cases (2–4 positive lymph nodes and ENE-negative) commonly receive surgery followed by radiotherapy. Accordingly, stepwise deterioration in survival was observed in these subgroups (appendix p 6).

AJCC9V consistently outperformed AJCC8E in different statistical aspects, including hazard consistency, outcome prediction (likelihood difference and explained variance), and balance (table 3). AJCC9V was inferior with regard to hazard discrimination, mainly due to the greater number of categories.

However, overall rank (a composite measure of the above factors) favoured AJCC9V. Regarding performance, AJCC9V showed an equivalent Brier Score to AJCC8E (0.041 vs 0.041) and a superior C-index (0.705 vs 0.702) in the validation cohort.

Discussion

In this study, we refined AJCC8E pathological staging to correct imbalances and reflect contemporary management. Using large-scale data with a derivation-validation

design, we confirmed that metastatic lymph node number along with ENE are the dominant nodal factors driving prognosis. By implementing statistical AHR and RPA models, N categorisation and TNM stage groupings were modified to integrate these determinants. Upon benchmarking with Groome's criteria, we confirmed that AJCC9V improved comparative performance.

It is anticipated that AJCC9V should improve hazard consistency within each stage, achieve greater balance between groups, and better align with treatment patterns.

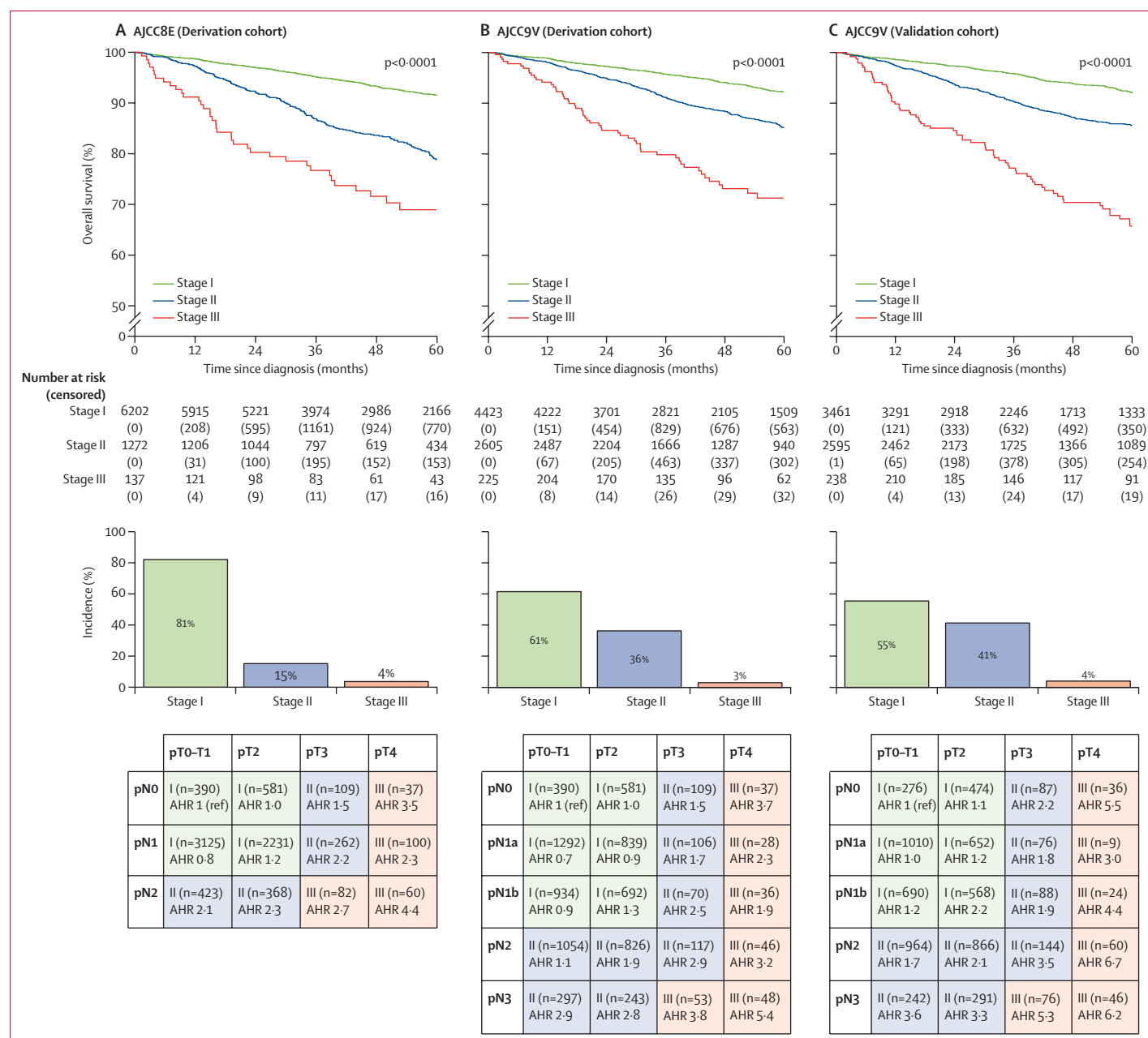


Figure 3: Generation of proposed AJCC9V pathological TNM staging for HPV-positive oropharyngeal carcinoma with associated Kaplan-Meier overall survival curves

(A) AJCC8E pTNM (derivation cohort). (B) AJCC9V pTNM (derivation cohort). (C) AJCC9V pTNM (validation cohort). AHR=adjusted hazard ratio. AJCC7E=American Joint Committee on Cancer staging system, 7th edition. AJCC8E=American Joint Committee on Cancer staging system, 8th edition. AJCC9V=American Joint Committee on Cancer staging system, version 9. HPV=human papillomavirus.

	Derivation cohort		Validation cohort		Combined cohort	
	AJCC8E	AJCC9V	AJCC8E	AJCC9V	AJCC8E	AJCC9V
Hazard consistency						
Score	2.81	2.65	2.84	2.15	4.92	4.15
Standardised score	1.00	0.00	1.00	0.00	1.00	0.00
Rank	2	1	2	1	2	1
Hazard discrimination						
Score	1.63	0.37	0.12	3.41	0.48	0.93
Standardised score	1.00	0.00	0.00	1.00	0.00	1.00
Rank	2	1	1	2	1	2
Likelihood difference						
Score	42.06	44.02	45.12	53.81	87.82	97.49
Standardised score	1.00	0.00	1.00	0.00	1.00	0.00
Rank	2	1	2	1	2	1
Explained variance						
Score	20.12	22.71	9.95	10.50	12.64	13.50
Standardised score	1.00	0.00	1.00	0.00	1.00	0.00
Rank	2	1	2	1	2	1
Balance						
Score	0.96	0.73	0.91	0.62	0.94	0.68
Standardised score	1.00	0.00	1.00	0.00	1.00	0.00
Rank	2	1	2	1	2	1
Overall rank						
Overall score	5.00	0.00	4.00	1.00	4.00	1.00
Rank	2	1	2	1	2	1
Performance						
Brier score	0.037	0.037	0.041	0.041	0.039	0.039
C-index	0.706	0.708	0.702	0.705	0.703	0.705

AJCC8E=American Joint Committee on Cancer staging system, 8th edition. AJCC9V=American Joint Committee on Cancer staging system, version 9.

Table 3: Groome's criteria comparison of AJCC8E and AJCC9V pathological TNM staging schema

In particular, AJCC8E N1 patients are now split across AJCC9V N1a, N1b, and N2. Such revision fittingly supports current practice, especially for AJCC9V N1a (1 positive lymph nodes and ENE-negative) as its own category. These patients comprise a sizeable percentage of cases, have a comparatively better prognosis than other AJCC8E N1 patients, and often receive single-modality treatment. The other N categories similarly reflect escalating mortality risk in stepwise fashion, rather than being lumped into one cluster. While the imbalance is not itself statistically erroneous, it confers less practical utility with so many patients in so few categories.¹⁴ The proposed allocation is clinically desirable by unmasking the heterogeneity within the dominant AJCC8E N1 cluster.

This sub-stratification in turn acknowledges the importance of ENE, bringing HPV-positive oropharyngeal carcinoma in line with other head and neck cancers. Previously termed extracapsular spread, ENE is regarded as a major risk factor meriting adjuvant chemoradiotherapy across head and neck cancers. It also is a key determinant for stratifying treatment groups in

numerous trials.^{5,6} However, ENE previously failed to meet criteria for inclusion in HPV-positive oropharyngeal carcinoma staging due to conflicting evidence.^{15–17}

Several factors could account for the divergent literature on ENE. ENE's absence of effect in earlier studies may stem from higher survival rates inherent to HPV-positive disease, which require greater numbers to power differences in outcome. By contrast, our large-scale analysis showed that ENE signals a clear mortality risk. Systemic therapy is moreover often given in ENE-positive cases, likely dampening its true magnitude of effect. It is further notable that surgical cohorts often exclude advanced or gross ENE cases, routing them to chemoradiotherapy; as such, our findings might in fact understate ENE's relative importance. Indeed, others have shown convincingly that radiological ENE as a correlate appears to worsen survival.¹⁸ All of these themes support the inclusion of ENE in staging.^{2,19,20}

The evolving use of ENE degree or extent further underscores its potential value to stratify risk. Previous work has suggested that ENE gradations in head and neck cancer might independently determine prognosis.¹⁹ The College of American Pathologists describes macroscopic pENE to be greater than 2 mm,²¹ while studies such as E3311 have defined higher-risk pENE thresholds to be greater than 1 mm.⁵ Our own analysis indicated no significant difference. However, pathological criteria for stratifying major and minor ENE are not well developed and might be prone to interrater reliability issues. With longer follow-up and refinement of pathological criteria,²² prognostic distinctions between ENE extent might become obvious and inform future AJCC iterations. The need for meaningful covariates such as ENE to assess treatment intensity continues to be highly relevant for HPV-positive disease, in which recurrence rates (10–25%)^{4,23} are substantial and outcomes for recurrent or metastatic disease (2-year overall survival 55%)²⁴ belie its perception as a low-risk cancer.

The final key change for AJCC9V involves rebalancing stage groups, which is crucial for trial design and prognostic relevance. Previous AJCC8E stages showed substantial disparities in incidence, with Stage I exceeding 80% of cases. Our AHR model corrects this top-heavy distribution: AJCC9V Stage I (55% of cases) and Stage II (41% of cases) have a far more uniform allocation. The system is also statistically intuitive, allowing T0-2N0-1 cases to remain Stage I. The stage groupings preserve Groome's proposed criteria for staging—specifically, superior hazard consistency, outcome prediction, balance, and performance. In exchange, hazard discrimination (separation between curves) is somewhat diminished. AJCC9V's concordance scores, although improved, nonetheless indicate imperfect performance and imply some chance of misclassification and overtreatment or undertreatment. This exemplifies what has been described as the

Goldilocks principle:^{14,25} the tradeoff between a perfect but complex model exploiting all possible determinants, and a pragmatic one based on usability.

It is worthwhile to note that as a consequence of redistribution, AJCC9V survival curves illustrate classic stage migration, also known as the Will Rogers phenomenon.²⁶ In effect, the worst AJCC8E Stage I cases are now distributed across advanced AJCC9V stages, becoming the best actors within their new stage group. As a result, all AJCC9V stage groups appear to improve when in fact nothing has changed. This has cautionary implications for judging the impact of interventions, and suggests that stage-to-stage comparisons between AJCC8E and AJCC9V should be avoided.

Several caveats deserve mention. Registry data are by nature retrospective, subject to coding error, and lack the granularity of institutional studies. Although this analysis involves the largest known HPV-positive oropharyngeal carcinoma cohort to date, it is based entirely on patients from the USA (the majority of whom were White and male) and might not be wholly generalisable to other countries or health-care settings. HPV testing was not uniform across institutions, and treatment protocols were not standardised. Conversely, the strength of registry data is how it reflects common practice rather than best practice. Our study also does not capture disease-specific survival; however, AJCC specifically focuses on overall survival as its primary endpoint, since this has fewer methodological confounders. The inclusion of ENE in staging also does not definitively describe its role as an indication for systemic therapy in HPV-positive oropharyngeal carcinoma.

Notably, p16 remains the established surrogate marker for high-risk, transcriptionally active HPV variants. Because p16 is broadly available worldwide, inexpensive, and correlates highly with HPV in the oropharynx, it is the principal assay used to diagnose HPV-positive oropharyngeal carcinoma. However, we are careful to acknowledge that p16 is not infallible: one international study showed that 10·9% of p16-positive patients were HPV-negative, while 7·5% of p16-negative patients were in fact HPV-positive.²⁷ Such discordant cases raise practical concerns, since their prognosis appears to be worse than that for p16-positive and HPV-positive cases, and their prevalence varies widely by geographical region. The College of American Pathologists now recommends additional HPV-specific testing beyond p16 in select circumstances, such as in regions with low HPV-positive oropharyngeal carcinoma prevalence or in cases with equivocal p16 staining.²⁸ As staging and treatment increasingly diverge in oropharyngeal carcinoma on the basis of HPV status and de-escalation strategies emerge, it is worthwhile to further characterise this phenomenon to avoid misclassification.

Finally, the development of circulating tumour HPV DNA has inspired proposals for diagnosis, staging, and

treatment, especially as a non-anatomical determinant. Although promising, circulating tumour HPV DNA testing is not yet universally available, and quantitative levels that indicate disease burden or prognosis are not yet established or harmonised. Furthermore, approximately 10% of confirmed HPV-positive oropharyngeal carcinoma cases have undetectable levels of circulating tumour HPV DNA,^{29,30} either because certain HPV subtypes are not covered or because detection thresholds are too high. Nonetheless, we expect that TNM staging systems across cancers will continue to evolve from anatomical factors to molecular ones, to augment outcome prediction. We await further study on circulating tumour HPV DNA to potentially inform future staging schema.

Altogether, these data collectively represent compelling evidence for staging modification. Constructing an optimised stage classification is a challenging endeavour, requiring balance between competing priorities. AJCC9V updates HPV-positive oropharyngeal carcinoma staging to reflect current practice, respects staging principles proposed by Groome and colleagues,¹² and implements changes that improve prognostication. Future work to independently validate the performance of this schema is warranted.

Contributors

ASH and ZSZ contributed to the conceptualisation of the study, data curation, formal analysis, methodology, validation, writing the original draft, and review and editing of the manuscript. SHH, JSL, HM, and NFS contributed to the conceptualisation of the study, the methodology, validation, and review and editing of the manuscript. BO and ML contributed to the formal analysis, methodology, validation, and review and editing of the manuscript. ME and RLF contributed to the conceptualisation of the study, formal analysis, validation, and review and editing of the manuscript. RRS, SSY, and BFB contributed to the conceptualisation of the study, validation, and review and editing of the manuscript. SGP and WML contributed to the conceptualisation of the study, formal analysis, methodology, validation, and review and editing of the manuscript. All authors had full access to all the data reported in the study, and had final responsibility for the decision to submit the manuscript for publication.

Declaration of interests

NFS reports having a compensated or non-compensated advisory role with AstraZeneca, Eisai Medical, Exelixis, Merck, Merck EMD Serono, Pfizer, Kura, Vaccinex, CUE, BionTech, GSK, TOSK, Seagen, Flamingo, Infinity, Inovio, Aveo, Medscape, Onclive, Uptodate, BMS, Cornerstone, Celldex, Surface Oncology, Urogen, Astex, Imugene, Faron Pharmaceutical, Coherus, Adagene, Fulgent, Springer, Nanobiotix, and Taiho. SGP reports the following patents: PCT/US2016/026717 ("Methods of Cancer Detection Using PARPI-FL"; pending); US 10,016,238 B2 ("Apparatus, system and method for providing laser steering and focusing for incision, excision and ablation of tissue in minimally-invasive surgery"); PCT/US2014/073053 ("Systems, methods, and apparatus for multichannel imaging of fluorescent sources in real time"); PCT/US2015/065816 ("Cyclic peptides with enhanced nerve-binding selectivity, nanoparticles bound with said cyclic peptides, and use of same for real-time in vivo nerve tissue imaging"); PCT/US2016/066969 ("Imaging systems and methods for tissue differentiation, e.g., for intraoperative visualization"); and WO2019212945A1 ("Nanotherapeutic systems and methods using particle-driven photodynamic therapy (PDT)"). RLF reports support from AstraZeneca/MedImmune, Merck, and Merus NV; consulting fees from Nanobiotix, Novasenta, and Novotech; advisory board participation with Bicara Therapeutics, Bobcat Bio, Coherus Biosciences, CoriRX Therapeutics, CureVac, Merck, Merus NV, Meira GTx, Mirror Biologics,

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Data sharing

NCDB data are non-confidential and publicly accessible. Data can be extracted through the American College of Surgeons website (<https://www.facs.org/quality-programs/cancer-programs/national-cancer-database/>).

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